

## CAPE SAN BLAS, FL, USA

WMO: 994360

Lat: 29.672N

Lon: 85.357W

Elev: 2

StdP: 101.30

Time Zone: -5.00 (NAE)

Period: 90-04

WBAN: 99999

## Annual Heating, Humidification, and Ventilation Design Conditions

| Coldest Month | Heating DB |            | Humidification DP/MCDB and HR |            |            |            |            |            | Coldest Month WS/MCDB |            |             |            | MCWS/PCWD<br>to 99.6% DB |           | WSF          |
|---------------|------------|------------|-------------------------------|------------|------------|------------|------------|------------|-----------------------|------------|-------------|------------|--------------------------|-----------|--------------|
|               |            |            | 99.6%                         |            |            | 99%        |            |            | 0.4%                  |            | 1%          |            |                          |           |              |
|               | 99.6%      | 99%        | DP                            | HR         | MCDB       | DP         | HR         | MCDB       | WS                    | MCDB       | WS          | MCDB       | MCWS                     | PCWD      |              |
| (a)<br>1      | (b)<br>2.8 | (c)<br>4.9 | (d)<br>N/A                    | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>15.9           | (k)<br>6.7 | (l)<br>14.6 | (m)<br>7.3 | (n)<br>5.0               | (o)<br>30 | (p)<br>0.420 |

## Annual Cooling, Dehumidification, and Enthalpy Design Conditions

| Hottest Month | Hottest Month DB Range | Cooling DB/MCWB |            |             |            |             |            | Evaporation WB/MCDB |            |            |            |            |            | MCWS/PCWD to 0.4% DB |            |
|---------------|------------------------|-----------------|------------|-------------|------------|-------------|------------|---------------------|------------|------------|------------|------------|------------|----------------------|------------|
|               |                        | 0.4%            |            | 1%          |            | 2%          |            | 0.4%                |            | 1%         |            | 2%         |            |                      |            |
|               |                        | DB              | MCWB       | DB          | MCWB       | DB          | MCWB       | WB                  | MCDB       | WB         | MCDB       | WB         | MCDB       | MCWS                 | PCWD       |
| (a)<br>8      | (b)<br>3.6             | (c)<br>30.2     | (d)<br>N/A | (e)<br>29.8 | (f)<br>N/A | (g)<br>29.4 | (h)<br>N/A | (i)<br>N/A          | (j)<br>N/A | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>3.9           | (p)<br>260 |

| Dehumidification DP/MCDB and HR |            |            |            |            |            |            |            |            | Enthalpy/MCDB |            |            |            |            |            | Extreme<br>Max WB |
|---------------------------------|------------|------------|------------|------------|------------|------------|------------|------------|---------------|------------|------------|------------|------------|------------|-------------------|
| 0.4%                            |            |            | 1%         |            |            | 2%         |            |            | 0.4%          |            | 1%         |            | 2%         |            |                   |
| DP                              | HR         | MCDB       | DP         | HR         | MCDB       | DP         | HR         | MCDB       | Enth          | MCDB       | Enth       | MCDB       | Enth       | MCDB       |                   |
| (a)<br>N/A                      | (b)<br>N/A | (c)<br>N/A | (d)<br>N/A | (e)<br>N/A | (f)<br>N/A | (g)<br>N/A | (h)<br>N/A | (i)<br>N/A | (j)<br>N/A    | (k)<br>N/A | (l)<br>N/A | (m)<br>N/A | (n)<br>N/A | (o)<br>N/A | (p)<br>N/A        |

## Extreme Annual Design Conditions

| Extreme Annual WS |          |         | Extreme Annual Temperature |         |                    |         | n-Year Return Period Values of Extreme Temperature |         |            |         |            |         |            |         |  |  |
|-------------------|----------|---------|----------------------------|---------|--------------------|---------|----------------------------------------------------|---------|------------|---------|------------|---------|------------|---------|--|--|
|                   |          |         | Mean                       |         | Standard Deviation |         | n=5 years                                          |         | n=10 years |         | n=20 years |         | n=50 years |         |  |  |
| 1%                | 2.5%     | 5%      | Min                        | Max     | Min                | Max     | Min                                                | Max     | Min        | Max     | Min        | Max     | Min        | Max     |  |  |
| (a) 13.1          | (b) 10.6 | (c) 9.0 | (e) N/A                    | (f) N/A | (g) N/A            | (h) N/A | (i) N/A                                            | (j) N/A | (k) N/A    | (l) N/A | (m) N/A    | (n) N/A | (o) N/A    | (p) N/A |  |  |
|                   |          |         | DB                         | N/A     | N/A                | N/A     | N/A                                                | N/A     | N/A        | N/A     | N/A        | N/A     | N/A        | N/A     |  |  |
|                   |          |         | WB                         | N/A     | N/A                | N/A     | N/A                                                | N/A     | N/A        | N/A     | N/A        | N/A     | N/A        | N/A     |  |  |

## Monthly Climatic Design Conditions

|                                                                        |             | Annual | Jan   | Feb   | Mar   | Apr   | May   | Jun   | Jul   | Aug   | Sep   | Oct   | Nov   | Dec  |  |  |
|------------------------------------------------------------------------|-------------|--------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|-------|------|--|--|
|                                                                        |             | (d)    | (e)   | (f)   | (g)   | (h)   | (i)   | (j)   | (k)   | (l)   | (m)   | (n)   | (o)   | (p)  |  |  |
| (6) Temperatures, Degree-Days and Degree-Hours                         | DBAvg       | 19.3   | 12.1  | 13.3  | 15.7  | 18.6  | 22.2  | 24.6  | 24.8  | 24.8  | 24.6  | 20.7  | 15.9  | 13.2 |  |  |
|                                                                        | DBStd       | 6.42   | 4.03  | 3.60  | 3.91  | 4.03  | 4.15  | 4.21  | 5.22  | 4.89  | 4.24  | 4.04  | 4.59  | 4.02 |  |  |
|                                                                        | HDD10.0     | 67     | 25    | 11    | 7     | 1     | 0     | 0     | 0     | 0     | 0     | 1     | 7     | 15   |  |  |
|                                                                        | HDD18.3     | 858    | 193   | 142   | 95    | 44    | 22    | 17    | 22    | 18    | 16    | 27    | 99    | 163  |  |  |
|                                                                        | CDD10.0     | 3444   | 92    | 104   | 183   | 260   | 379   | 438   | 459   | 460   | 438   | 332   | 184   | 116  |  |  |
|                                                                        | CDD18.3     | 1193   | 1     | 1     | 12    | 54    | 144   | 204   | 222   | 220   | 204   | 99    | 27    | 5    |  |  |
|                                                                        | CDH23.3     | 10617  | 0     | 0     | 1     | 74    | 835   | 2038  | 2626  | 2509  | 1998  | 488   | 47    | 0    |  |  |
| (14) Wind                                                              | WSAvg       | 4.0    | 4.1   | 4.5   | 4.4   | 4.3   | 4.2   | 4.2   | 3.9   | 3.6   | 3.5   | 3.6   | 3.9   | 4.0  |  |  |
|                                                                        | PrecAvg     | 1404   | 111   | 110   | 127   | 83    | 53    | 150   | 177   | 190   | 151   | 83    | 87    | 86   |  |  |
|                                                                        | PrecMax     | 2027   | 474   | 261   | 279   | 251   | 257   | 499   | 339   | 349   | 514   | 268   | 331   | 216  |  |  |
|                                                                        | PrecMin     | 876    | 15    | 9     | 0     | 8     | 0     | 10    | 81    | 57    | 2     | 0     | 4     | 37   |  |  |
|                                                                        | PrecStd     | 280    | 87    | 66    | 73    | 64    | 50    | 97    | 72    | 78    | 102   | 68    | 66    | 42   |  |  |
|                                                                        | DB          | 20.7   | 20.9  | 23.2  | 26.6  | 29.0  | 30.0  | 30.8  | 30.8  | 30.4  | 30.4  | 28.8  | 25.5  | 22.1 |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
| (20) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures | 2% DB       | 19.6   | 19.8  | 21.9  | 25.2  | 27.8  | 29.1  | 30.1  | 30.2  | 29.8  | 28.0  | 24.6  | 21.1  |      |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | 5% DB       | 18.6   | 19.0  | 21.1  | 24.1  | 27.0  | 28.7  | 29.6  | 29.8  | 29.2  | 27.0  | 23.8  | 20.2  |      |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | 10% DB      | 17.6   | 18.3  | 20.4  | 23.3  | 26.5  | 28.3  | 29.3  | 29.4  | 28.9  | 26.0  | 22.4  | 19.1  |      |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
| (28) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures | 0.4% WB     | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | 2% WB       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | 5% WB       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | MCDB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | 10% WB      | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
| (35) Mean Daily Temperature Range                                      | MDBR        | 5.5    | 5.3   | 4.7   | 4.3   | 3.8   | 3.7   | 3.2   | 3.6   | 4.5   | 5.2   | 6.2   | 5.6   |      |  |  |
|                                                                        | 5% DB       | 5.2    | 5.1   | 4.4   | 4.6   | 4.6   | 4.7   | 3.9   | 4.3   | 5.1   | 5.1   | 5.4   | 5.1   |      |  |  |
|                                                                        | MCWBR       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | 5% WB       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | MCDBR       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | MCWBR       | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
|                                                                        | MCWB        | N/A    | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A   | N/A  |  |  |
| (40) Clear-Sky Solar Irradiance                                        | taub        | 0.349  | 0.358 | 0.369 | 0.396 | 0.434 | 0.462 | 0.487 | 0.481 | 0.434 | 0.389 | 0.358 | 0.348 |      |  |  |
|                                                                        | taud        | 2.513  | 2.495 | 2.466 | 2.387 | 2.313 | 2.292 | 2.239 | 2.263 | 2.367 | 2.478 | 2.496 | 2.534 |      |  |  |
|                                                                        | Ebn at Noon | 886    | 908   | 918   | 899   | 861   | 832   | 810   | 812   | 842   | 864   | 867   | 869   |      |  |  |
|                                                                        | Edh at Noon | 90     | 99    | 108   | 121   | 131   | 133   | 140   | 136   | 118   | 100   | 90    | 84    |      |  |  |
| (44) All-Sky Solar Radiation                                           | RadAvg      | 3.17   | 3.85  | 5.14  | 6.15  | 6.86  | 6.32  | 5.89  | 5.49  | 5.09  | 4.57  | 3.57  | 2.84  |      |  |  |
|                                                                        | RadStd      | 0.26   | 0.47  | 0.42  | 0.51  | 0.42  | 0.43  | 0.41  | 0.38  | 0.34  | 0.33  | 0.27  | 0.26  |      |  |  |

## Historical Trends

|                              |  | DBAvg | Heating |        | Cooling |       |       | Degree-Days |         |         |         |
|------------------------------|--|-------|---------|--------|---------|-------|-------|-------------|---------|---------|---------|
|                              |  |       | 99% DB  | 99% DP | 1% DB   | 1% WB | 1% DP | HDD10.0     | HDD18.3 | CDD10.0 | CDD18.3 |
| (46) Station Only            |  | N/A   | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | N/A     | N/A     | N/A     |
| (47) Regional (27 neighbors) |  | +0.45 | N/A     | N/A    | N/A     | N/A   | N/A   | N/A         | -88     | +154    | +100    |

Nomenclature: See separate page